

# CURRICULUM VITAE

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## RESEARCH PROFILE

Researcher specializing in computational imaging and inverse problems, with a particular focus on computed tomography (CT) image reconstruction from incomplete and limited measurement data. Research experience spans compressed sensing, limited-angle and sparse-view tomography, level-set methods, physics-guided deep learning, generative models, and hybrid reconstruction methods. Current research focuses on developing advanced mathematical and learning-based approaches for solving challenging CT image reconstruction problems.

## RESEARCH INTERESTS

- Computational imaging and inverse problems
- CT image reconstruction
- Compressed sensing
- Numerical optimization for image reconstruction
- Generative models for inverse problems

## EDUCATION

| Start and end dates |    |   |      |    | Name of institution, school, department, major                                                                                                                          | Type of Completion |
|---------------------|----|---|------|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|
| YYYY                | MM | ~ | YYYY | MM |                                                                                                                                                                         |                    |
| 2006                | 9  | ~ | 2009 | 7  | Gizera High School                                                                                                                                                      | Graduation         |
| 2009                | 9  | ~ | 2013 | 7  | Sohag University, Faculty of Science, Department of Mathematics and Computer Science, bachelor's degree (Egypt)                                                         | Graduation         |
| 2016                | 4  | ~ | 2018 | 9  | Sohag University, Faculty of Science, Department of Mathematics and Computer Science, Master's degree (Egypt)                                                           | Completion         |
| 2020                | 10 | ~ | 2023 | 9  | University of Tsukuba, Graduate School of Science and Technology, Systems and Information Engineering, Degree Program in Computer Science, Ph.D. in Engineering (Japan) | Completion         |

## WORK EXPERIENCE

| Start and end dates |    |   |      |    | Employer, position                                                                             | Job description                                                                                        | Work Situation             |
|---------------------|----|---|------|----|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------|
| YYYY                | MM | ~ | YYYY | MM |                                                                                                |                                                                                                        |                            |
| 2016                | 4  | ~ | 2018 | 8  | Department of Mathematics and Computer Science, Sohag University.<br><b>Research Assistant</b> | Assisted students with exercises and guided them during tutorials.                                     | Full time<br>35 hours/week |
| 2018                | 9  | ~ | 2020 | 9  | Department of Mathematics and Computer Science, Sohag University.<br><b>Assistant Lecturer</b> | Prepared lecture materials, graded assignments and examinations, and guided students during tutorials. | Full time<br>35 hours/week |

## WORK EXPERIENCE — continued

| Start and end dates |    |   |      |    | Employer, position                                                                                        | Job description                                                                                        | Work Situation             |
|---------------------|----|---|------|----|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------|
| YYYY                | MM | ~ | YYYY | MM |                                                                                                           |                                                                                                        |                            |
| 2021                | 7  | ~ | 2022 | 3  | Systems and Information Engineering<br>University of Tsukuba.<br><b>Teaching Assistant (TA)</b>           | Organizing and preparing seminars, monitoring attendance, and mentoring students.                      | Part time<br>3 hours/week  |
| 2022                | 7  | ~ | 2023 | 3  | Systems and Information Engineering<br>University of Tsukuba.<br><b>Teaching Assistant (TA)</b>           | Organizing and preparing seminars, monitoring attendance, and mentoring students.                      | Part time<br>3 hours/week  |
| 2023                | 10 | ~ | 2024 | 2  | Department of Mathematics and Computer Science,<br>Sohag University.<br><b>Assistant Lecturer</b>         | Prepared lecture materials, graded assignments and examinations, and guided students during tutorials. | Full time<br>35 hours/week |
| 2024                | 3  | ~ | 2024 | 6  | Department of Mathematics and Computer Science,<br>Sohag University.<br><b>Lecturer</b>                   | Delivered lectures, prepared course materials, graded examinations, and guided students.               | Full time<br>35 hours/week |
| 2024                | 7  | ~ | 2026 | 7  | Systems and Information Engineering<br>University of Tsukuba.<br><b>JSPS Postdoctoral Research Fellow</b> | Conducted research on physics-guided deep learning methods for CT image reconstruction.                | Full time<br>40 hours/week |
| 2026                | 8  | ~ | now  |    | Systems and Information Engineering<br>University of Tsukuba.<br><b>Researcher</b>                        | Conducting research on CT image reconstruction using hybrid models.                                    | Full time<br>40 hours/week |

## HONORS AND AWARDS

| Date | Description                                                                                                                                  |
|------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | Encouragement Award, 45th Annual Meeting of the Japanese Society of Medical Imaging Technology (JAMIT), Tokyo, Japan.                        |
| 2026 | Best Poster Award, International Symposium on Biomedical Imaging (ISBI), London, UK.                                                         |
| 2025 | Best Presentation Award, International Conference on Computer and Automation Engineering (ICCAE), Perth, Australia.                          |
| 2024 | JSPS Research Fellowship, a prestigious fellowship awarded by the Japan Society for the Promotion of Science (JSPS) for research excellence. |
| 2023 | Academic Group Chair Award (研究群長表彰), University of Tsukuba (2023), for outstanding research achievements during doctoral studies.            |
| 2023 | Excellent Oral Presentation Award, International Conference on Computer and Automation Engineering (ICCAE), Sydney, Australia.               |

## FUNDING AND GRANTS

| Date | Description                                                                                                                                                         |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2024 | Grant-in-Aid for JSPS Fellows (KAKENHI Project Number: 24KF0101), University of Tsukuba, Japan, Co-Principal Investigator (July 2024 – July 2026) (1,900,000 yen).  |
| 2023 | Travel Grant, RIMS Workshop on Inverse Problems, Medical Imaging, and Related Topics, Kyoto, Japan, (70,000 yen).                                                   |
| 2022 | NEC C&C Foundation's grant for researchers, University of Tsukuba, Japan, Principal Investigator (Oct. 2022 – Mar. 2023) (200,000 yen).                             |
| 2020 | Academic of Scientific Research & Technology research grant, Sohag University, Egypt, Co-Principal Investigator (April. 2020 – Mar. 2021) (100,000 Egyptian Pound). |

## PROFESSIONAL AFFILIATIONS AND MEMBERSHIPS

|                 |                                                        |
|-----------------|--------------------------------------------------------|
| Name of society | Japanese Society of Medical Imaging Technology (JAMIT) |
|-----------------|--------------------------------------------------------|

## LIST OF SELECTED PUBLICATIONS AND PRESENTATIONS

### (1) Peer-reviewed journal papers

1. Haytham A. Ali and Hiroyuki Kudo. "PIDA-GAN: Physics-Informed Dual-Stage Attention GAN with Data Consistency for Sparse-View CT Reconstruction" *Physics in Medicine & Biology*, vol. 71, no. 15, Paper No. 155003, 2026.
2. Haytham A. Ali, Essam A. Rashed, and Hiroyuki Kudo. "Compressed sensing-based image reconstruction for discrete tomography with sparse view and limited angle geometries" *PLOS ONE*, vol. 20, no. 7, Paper No. e0327666, 2025.
3. Haytham A. Ali and Hiroyuki Kudo. "Binary Tomography Reconstruction with Limited-Data by a Convex Level-Set Method" *Computers, Materials & Continua*, vol. 73, no.2, pp. 3741-3756, 2022.
4. Abdel-Aziz, H. S., Zanaty, E. A., Haytham A. Ali, & Saad, M. K. "Generating Bézier curves for medical image reconstruction" *Results in Physics*, vol. 23, Paper No. 103996, 2021.

### (2) Peer-reviewed international conference papers

1. Haytham A. Ali and Hiroyuki Kudo. "Multi-Frequency and Locality-Guided Attention U-Net GAN for Limited-Angle CT Reconstruction" *IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, London, United Kingdom, pp. 1-4, 2026.
2. Haytham A. Ali and Hiroyuki Kudo. "PCDS-GAN: Physics-Constrained Dual-Stage Generative Model with Adaptive Gating for Ultra Sparse-View CT Reconstruction" *IEEE 23rd International Symposium on Biomedical Imaging (ISBI)*, London, United Kingdom, pp. 1-4, 2026.
3. Haytham A. Ali and Hiroyuki Kudo. "FLoAT-GAN: Frequency- and Locality-Guided Attention for Limited-Angle CT Reconstruction" *International Workshop on Health Intelligence (W3PHIAI-26)*, AAAI-26 workshop, Singapore, 2026, pp. 1-8. (Accepted for publication).
4. Haytham A. Ali and Hiroyuki Kudo. "Attention-guided frequency and locality learning GAN for limited-angle CT reconstruction" In *International Workshop on Advanced Imaging Technology (IWAIT)*, vol. 14072, pp. 596-601. SPIE, 2026.
5. Haytham A. Ali and Hiroyuki Kudo. "An Attention-Based Generative Adversarial Network for Limited-Angle CT Reconstruction" *17th International Conference on Computer and Automation Engineering (ICCAE)*, Perth, Australia, pp. 110-114, 2025.
6. Haytham A. Ali and Hiroyuki Kudo. "Level-Set Method for Limited-Data Reconstruction in CT using Dictionary-Based Compressed Sensing" *15th International Conference on Computer and Automation Engineering (ICCAE)*, Sydney, Australia, pp. 264-268, 2023.
7. Haytham A. Ali and Hiroyuki Kudo. "New Level-Set-Based Shape Recovery Method and its application to sparse-view shape tomography" In *Proceedings of the 4th International Conference on Digital Medicine and Image Processing (DMIP '21)*. Association for Computing Machinery, New York, NY, USA, pp. 24–29, 2021.

### **(3) Domestic conference presentations**

1. Haytham A. Ali, Hiroyuki Kudo "Recipe for Deep Learning Image Reconstruction for Ultra-Sparse-View CT" Annual Meeting of the Japanese Society of Medical Imaging Technology (JAMIT), Nagoya, Japan, 2026.
2. Haytham A. Ali, Hiroyuki Kudo "Deblurring the Angle: Proposal of Realistic Model for Sparse-View CT and Sinogram Space Deep Learning Approach" Annual Meeting of the Japanese Society of Medical Imaging Technology (JAMIT), Nagoya, Japan, 2026.
3. Haytham A. Ali, Hiroyuki Kudo "Novel Dual-Stage GAN for High-Fidelity CT Image Reconstruction from Sparse view CT" Annual Meeting of the Japanese Society of Medical Imaging Technology (JAMIT), Tokyo, Japan, 2025.
4. Haytham A. Ali, Hiroyuki Kudo "Self-Attention GAN for High-Quality Limited-Angle CT Reconstruction" Annual Meeting of the Japanese Society of Medical Imaging Technology (JAMIT), Tokyo, Japan, 2025.
5. Haytham A. Ali, Katsuya Fujii, and Hiroyuki Kudo "Reconstruction method for binary limited data using a dictionary-based sparse shape recovery" Annual Meeting of the Japanese Society of Medical Imaging Technology (JAMIT), Nagoya, Japan, 2022.

### **(4) Invited Talks**

1. Haytham A. Ali "Learning-Based Approaches for CT Reconstruction with Incomplete Projection Data" International Conference on Control and Computer Vision (ICCCV), Tsukuba, Japan, 2026.
2. Haytham A. Ali "Towards Accurate Reconstruction from Sparse and Limited-Angle CT Data Using Deep Learning" Asia Symposium on Image Processing (ASIP), Tsukuba, Japan, 2025.
3. Haytham A. Ali "Parametric Level Set Methods for Limited Data CT Image Reconstruction" Workshop on Inverse Problems, Medical Imaging, and Related Topics, Kyoto, Japan, 2023.

## **REFERENCES**

1. Prof. Hiroyuki Kudo  
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2. Prof. Essam A. Rashed  
Graduate School of Information Science  
University of Hyogo, Japan  
Email: [rashed@gsis.u-hyogo.ac.jp](mailto:rashed@gsis.u-hyogo.ac.jp)
3. Prof. Dr. M. Khalifa Saad  
Sohag University, Egypt  
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